S16 Governance Gate Ablation – Methodological Note

Purpose

This supplementary section describes how the sensitivity of the governance-gate architecture can be assessed through gate ablation and clarifies the epistemic limitations associated with such analyses within the HIMS-CDI framework.

Method

The full deployment label within the synthetic evaluation corpus is defined as:

DEPLOY_SYNTH = (CDI $\geq$ 0.70) $\wedge$ (CCR $\geq$ 0.80) $\wedge$ (latency < 200 ms) $\wedge$ (memory < 100 MB)

Removing any single governance gate would fundamentally alter the underlying ground-truth definition itself. Consequently, a conventional machine-learning ablation strategy based on comparing classification performance before and after removing a component is not directly applicable, because the positive class would no longer represent the same governance construct.

Accordingly, a principled gate-level sensitivity analysis should instead examine the following:

1. Gate-Level Pass Rates

For each of the 240 synthetic environments, the pass/fail status of every governance gate should be recorded individually.

2. Impact on Deployment Labels

The analysis should quantify how many environments would transition from NOT_DEPLOY_SYNTH to DEPLOY_SYNTH, and vice versa, if a particular governance gate were ignored.

3. Threshold Perturbation Analysis

Each governance threshold may be perturbed systematically. For example:

- CDI $\in$ [0.65, 0.75]
- CCR $\in$ [0.75, 0.85]
- Latency $\in$ [150 ms, 250 ms]
- Memory $\in$ [80 MB, 120 MB]

The resulting number of deployment-label flips can then be measured.

This methodology identifies the most restrictive governance constraints, and therefore those that exert the greatest influence on the deployment boundary.

Implementation Note

The complete governance-gate pass/fail records for the 240-environment synthetic corpus are archived in the project repository on GitHub (https://github.com/oluseun-akeredolu/HIMS-CDI). Researchers may regenerate all ablation statistics using the script:

analysis/gate_ablation.py

Because the manuscript’s evaluation test set contains only 48 environments, gate-level pass counts are necessarily small. Consequently, full-corpus analyses provide more statistically stable sensitivity estimates.

Conclusion

Governance-gate ablation within HIMS-CDI should be interpreted as a structural sensitivity analysis rather than a classification-performance experiment. The proposed methodology avoids the methodological error of comparing models against a changing ground truth and instead provides a principled mechanism for assessing the contribution of each governance constraint to overall framework restrictiveness.